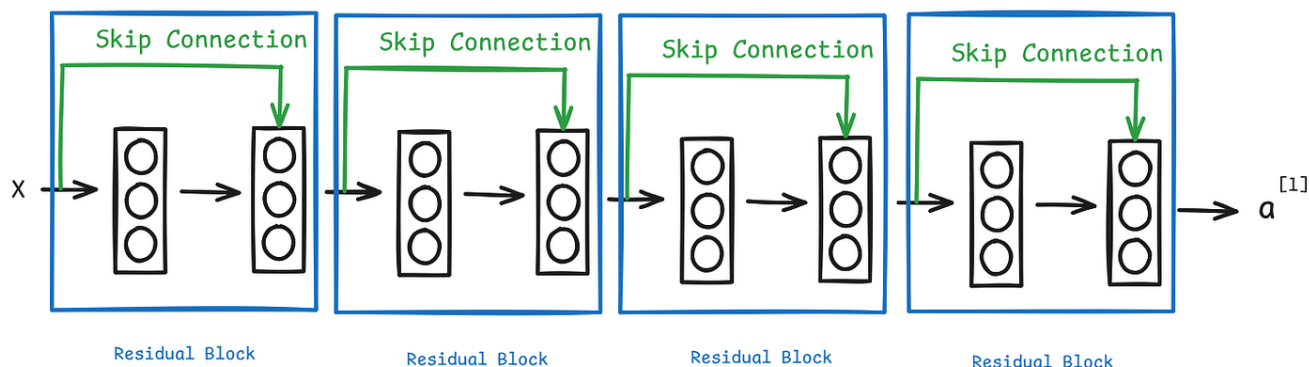


# Residual Connection - ResNets

Residual Connection is a DL technique that "Skip" one or more layers in a NN by creating a direct shortcut path.



It revolutionized the ML by solving the vanishing gradient problem, which previously made it nearly impossible to train very deep networks.

Without the shortcuts, gradients shrink to zero as they travel backward through dozens of layers, preventing the model from learning.

**Key benefits include:**

- **Unimpeded Gradient Flow:** Shortcuts provide a direct, unobstructed path for gradients to flow backward during training.
- **Easier Optimization:** Layers only need to learn the "residual" (the difference) between the input and the output, making the network's optimization task much simpler.
- **Deeper Architectures:** They allow models to have hundreds or thousands of layers without losing the ability to learn

It is used to mitigate a problem called "The degradation" in deep learning where adding more layers to a neural network causes its accuracy to saturate and then degrade rapidly